Roll Number:_____

Thapar Institute of Engineering & Technology

Department of Computer Science and Engineering

MID SEMESTER EXAMINATION

B. E. (2nd Yr. COE/CSE)
19th Mar, 2026
Time: 11:30 am to 1:30 pm
Max Marks: 40 (Weightage: 30)

Course Code: UCS310
Course Name: Database Management System
Name of Faculty: Dr. Simranjit (C6), Dr. Shashank (C3), Dr. Amrita (C1), Dr. Mahak (Q1, Q2), Dr. Shivani (C8, C5, Q4), Dr. Ananya (Q3, C7), Dr. Tanya (C2, C4), Dr. Manisha (P1)

Note: Attempt all Questions in sequence. Answer all sub-parts of each question at one place. Do mention Page No. of your attempt at front page of your answer sheet. Assume missing data (if any).

Q.No	Questions	MM	CO	BL
Q.1	**(a)** A school information system stores data about students, classes, and teachers. Discuss how entity integrity and referential integrity together help the system function correctly for both administrative staff and technical operations.	(3)		
	(b) Fill in blanks with most suitable answer. i) When changes in the conceptual schema do not affect user views, the system supports _____ data independence. ii) In an organization, different users such as managers, clerks, and analysts access different views of the same database; this reflects the role of the _____ level in supporting user-specific interaction. iii) The _____ level acts as a bridge between organizational data requirements and technical database design, thereby supporting both social and technical aspects of an information system.	(3)	CO1	L2
	(c) A university maintained student, course, exam, and fee records using separate files. As data increased, it became difficult to update records, avoid duplication, manage simultaneous access, and maintain accuracy. The university then adopted a Database Management System (DBMS). Compare the File-Based System and DBMS based on: i) Data Consistency ii) Data Redundancy iii) Concurrency Handling and iv) Data Integrity.	(4)		
Q.2	The CityHealth Hospital in Bengaluru is transitioning from a legacy database system to a centralized Hospital Information Management System to enhance efficiency and support sustainable development in healthcare through improved data management and resource optimization. You have been hired to gather system requirements and design the database for this new system. As part of the task, you are required to develop a conceptual ER model and map it to a minimized relational schema, while adhering to the following three rules. I. The hospital is divided into several **Wards (Ward_Number, Capacity).** Each Ward must be managed by **exactly one** Nurse acting as the **In_charge** Nurse. **Nurses** are identified by **Staff_ID, Name,** and **Shift_Type.** While every Ward requires a **In_charge** Nurse, only a small fraction of the nursing staff are **In_charge** Nurses. A Patient is admitted to a specific Ward. II. The hospital registers **Patients (Patient_ID, Name, and Blood_Group).** Every Patient must have **at least one** Attendant recorded. An attendant is not treated as an independent person in the system. They are only identified by their Relationship (e.g. Spouse, Sibling etc.). If a patient's record is deleted, their Attendant data must also be deleted. III. **Doctors (Doc_ID, Name, and Specialty)** treat multiple Patients. Similarly, a Patient with a complex illness will also be treated by multiple Doctors during their hospital visit. Whenever a doctor sees a Patient, the hospital billing system must record the specific **Consult_Date** and the **resulting Diagnosis** for that exact visit. **(a)** Draw the complete ER Diagram for the CityHealth system with all the cardinality ratios and participation constraints.	(10)	CO2	L4

	(b) Map the ER Diagram into the minimum possible number of relational tables. Determine the primary key and foreign key of each table after minimization.			
Q.3	A hospital is developing a digital health information system to monitor patient treatments, doctors, nurses, and ward utilization so that quality healthcare can be delivered efficiently and resources can be managed responsibly. To support this system, the hospital maintains treatment-related information in the following relation: **TREATMENT** (PatientID, PatientName, DoctorID, DoctorName, Specialization, WardNo, WardType, TreatmentID, TreatmentName, Cost, Date, NurseID, NurseName) The following functional dependencies are known: **PatientID → PatientName,** **DoctorID → DoctorName, Specialization,** **WardNo → WardType,** **TreatmentID → TreatmentName, Cost,** **NurseID → NurseName,** **(PatientID, TreatmentID, Date) → DoctorID, WardNo, NurseID** **(a)** Determine all candidate keys of the relation with proper justification. **(b)** Identify and explain the partial and transitive dependencies present in the schema. **(c)** Normalize the relation up to Third Normal Form (3NF), clearly showing each decomposition step. **(d)** Examine whether the resulting relations satisfy Boyce-Codd Normal Form (BCNF). If violations exist, perform appropriate decomposition.	(10)	CO3	L3
Q.4	**(a)** An eco-friendly retail company is developing a data management system to monitor product sales through both online platforms and physical stores. The goal is to analyze product demand, reduce overproduction, and manage inventory efficiently to minimize waste and promote sustainable retail practices. The company stores product and order information in the following relational schema: **Products** (ProductID, ProductName, CategoryID, Price), **OnlineOrders** (OrderID, ProductID) and **StoreOrders** (OrderID, ProductID). Write SQL queries to fetch following details from database: i) Display CategoryID and the number of products in each category that contains more than four products. ii) List all products and their online order IDs, including products with no online orders. iii) Show product IDs and order IDs from OnlineOrders and StoreOrders such that the result includes products appearing only in online orders as well as those appearing only in store orders. iv) Display ProductID, ProductName, Online OrderID, and Store OrderID for products that have been ordered both online and in-store. v) Display products that have been ordered either online or in-store, and whose price is greater than the average price of products in the same category.	(7)	CO5	L3
	(b) Suppose we have a database for an investment firm, consisting of the following attributes: B – Broker, O – Office of a broker, I – Investor, S – Stock, Q – Quantity of stock owned by an investor, D – dividend paid by a stock. Hence, the overall schema is R = (B, O, I, S, Q, D). Assume that the following functional dependencies are required to hold this database. **I → B, IS → Q, B → O, S → D** On the basis of given information, write true or false for the following statements: i) If R is decomposed into R1(I, B), R2(B, O), and R3(I, S, Q, D), the decomposition is lossless. ii) If R is decomposed into R1(I, B, O) and R2(I, S, Q, D), the decomposition is lossless but not dependency preserving. iii) If a decomposition preserves all functional dependencies, it automatically guarantees a lossless join.	(3)	CO3	L2